গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
সিলেট-৩১১৪, বাংলাদেশ।
যোগাযোগঃ +৮৮ ০১৩১৮ ২১০ ২৮৬/২৮৭/২৮৮/২৮৯/২৯০
Website: www.crtc-cee-sust.com

Center for Research, Testing and Consultancy
Department of Civil and Environmental Engineering
Shahjalal University of Science and Technology
Sylhet-3114, Bangladesh.
Contact: +88 01318 210 286/ 287/ 288/ 289/ 290
Email: crtc-cee@sust.edu

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SUB-SOIL INVESTIGATION REPORT

FOR THE PROPOSED
CONSTRUCTION OF 05 STORIED RESIDENTIAL BUILDING
AT EVERGREEN CITY, MIRAPARA, SHAHPORAN, SADAR, SYLHET.
DAG # SA-1427, BS-5333, J.L # SA-96, BS-81, MOUJA- DEVPUR.

CLIENT

SYED GOLAM AMBIA.

Verification Code: 5m461Gy6
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTED BY

M/S RELATION SOIL INVESTIGATION
Kumarpara, Sylhet.
Mob : 01711063018 & 01712186249.

Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-60 ft	12
02	Disturbed sample	02	0-60 ft	12
03	Disturbed sample	03	0-60 ft	12
04	Disturbed sample	04	0-65 ft	13
05	Disturbed sample	05	0-65 ft	13
06	Disturbed sample	06	0-65 ft	13
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

<u>Bearing capacity for deep foundation</u>

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a = Allowable bearing capacity, q_u = Unconfined Compressive Strength, γ = Unit Weight of soil, D = Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult} = Ultimate bearing capacity in kPa, N = SPT Number, FS = Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation = 9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Tables 7-11**.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS = 2.5)

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square (Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	8	0.928	0.976	1.064	1.096	1.184	1.288	1.000
	5.0-10	11	1.304	1.371	1.495	1.540	1.664	1.810	1.405
BH02	2.5-5.0	6	0.696	0.732	0.798	0.822	0.888	0.966	0.750
	5.0-10	11	1.272	1.338	1.458	1.502	1.623	1.765	1.371
BH03	2.5-5.0	8	0.882	0.927	1.011	1.041	1.125	1.224	0.950
	5.0-10	11	1.224	1.287	1.403	1.446	1.562	1.699	1.319
BH04	2.5-5.0	8	0.882	0.927	1.011	1.041	1.125	1.224	0.950
	5.0-10	12	1.420	1.493	1.628	1.677	1.812	1.971	1.530
BH05	2.5-5.0	7	0.835	0.878	0.958	0.986	1.066	1.159	0.900
	5.0-10	10	1.148	1.207	1.316	1.356	1.465	1.593	1.237
BH06	2.5-5.0	6	0.696	0.732	0.798	0.822	0.888	0.966	0.750
	5.0-10	9	1.020	1.073	1.169	1.205	1.301	1.416	1.099
BH07	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.032	0.032	0.096	0.096	0.049	0.096	-	-	-	-
10	0.096	0.049	0.096	0.118	0.074	0.118	-	-	-	-
15	0.112	0.088	0.104	0.096	0.112	0.174	-	-	-	-
20	0.120	0.152	0.112	0.152	0.096	0.104	-	-	-	-
25	0.074	0.156	0.096	0.138	0.118	0.138	-	-	-	-
30	0.232	0.232	0.232	0.232	0.232	0.208	-	-	-	-
35	0.232	0.096	0.240	0.232	0.240	0.224	-	-	-	-
40	0.246	0.118	0.246	0.230	0.261	0.239	-	-	-	-
45	0.096	0.138	0.074	0.246	0.246	0.226	-	-	-	-
50	0.138	0.246	0.118	0.401	0.281	0.241	-	-	-	-
55	0.669	0.669	0.192	0.669	0.375	0.522	-	-	-	-
60	0.184	0.192	0.192	0.200	0.184	0.184	-	-	-	-
65	-	-	-	0.184	0.200	0.192	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	6.400	6.400	0.900	0.900	0.450	0.900	-	-	-	-
10	0.900	0.450	0.900	1.125	0.675	1.125	-	-	-	-
15	22.400	17.600	20.800	19.200	22.400	1.800	-	-	-	-
20	24.000	30.400	22.400	30.400	19.200	20.800	-	-	-	-
25	0.675	1.575	0.900	1.350	1.125	1.350	-	-	-	-
30	46.400	46.400	46.400	46.400	46.400	41.600	-	-	-	-
35	46.400	0.900	3.825	46.400	48.000	44.800	-	-	-	-
40	3.375	1.125	3.375	3.150	2.700	2.475	-	-	-	-
45	0.900	1.350	0.675	3.375	3.375	3.600	-	-	-	-
50	1.350	3.375	1.125	6.750	4.725	4.050	-	-	-	-
55	11.250	11.250	38.400	11.250	6.300	8.775	-	-	-	-
60	36.800	38.400	38.400	40.000	36.800	36.800	-	-	-	-
65	-	-	-	36.800	40.000	38.400	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.021	0.021	0.062	0.062	0.032	0.062	-	-	-	-
10	0.062	0.032	0.062	0.077	0.048	0.077	-	-	-	-
15	0.073	0.057	0.068	0.062	0.073	0.113	-	-	-	-
20	0.078	0.099	0.073	0.099	0.062	0.068	-	-	-	-
25	0.048	0.101	0.062	0.090	0.077	0.090	-	-	-	-
30	0.151	0.151	0.151	0.151	0.151	0.135	-	-	-	-
35	0.151	0.062	0.156	0.151	0.156	0.146	-	-	-	-
40	0.160	0.077	0.160	0.149	0.170	0.156	-	-	-	-
45	0.062	0.090	0.048	0.160	0.160	0.147	-	-	-	-
50	0.090	0.160	0.077	0.261	0.183	0.156	-	-	-	-
55	0.435	0.435	0.125	0.435	0.243	0.339	-	-	-	-
60	0.120	0.125	0.125	0.130	0.120	0.120	-	-	-	-
65	-	-	-	0.120	0.130	0.125	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	2.880	2.880	0.405	0.405	0.203	0.405	-	-	-	-
10	0.405	0.203	0.405	0.506	0.304	0.506	-	-	-	-
15	10.080	7.920	9.360	8.640	10.080	0.810	-	-	-	-
20	10.800	13.680	10.080	13.680	8.640	9.360	-	-	-	-
25	0.304	0.709	0.405	0.608	0.506	0.608	-	-	-	-
30	20.880	20.880	20.880	20.880	20.880	18.720	-	-	-	-
35	20.880	0.405	1.721	20.880	21.600	20.160	-	-	-	-
40	1.519	0.506	1.519	1.418	1.215	1.114	-	-	-	-
45	0.405	0.608	0.304	1.519	1.519	1.620	-	-	-	-
50	0.608	1.519	0.506	3.038	2.126	1.823	-	-	-	-
55	5.063	5.063	17.280	5.063	2.835	3.949	-	-	-	-
60	16.560	17.280	17.280	18.000	16.560	16.560	-	-	-	-
65	-	-	-	16.560	18.000	17.280	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the client, following recommendations are suggested for the proposed five-storied residential building.

Soil profile scenario

Overall soil information of the investigated site is almost regular in between the boreholes and huge vertical inconsistency exists in the soil formation. The topmost 5-ft soil in boreholes 1 and 2 is loose grey coarse sand, followed by grey soft clay with very low bearing capacity. While boreholes 3-6 consist of grey/brown soft clay with very low bearing capacity up to 10 ft from EGL. After that 1/2 layers of fine sand have been observed in all the borehole locations, followed by a grey soft clay layer. An organic soil layer has been found at 25 ft depth in BH2. One or two layers of grey fine sand with good bearing capacity have been observed below 30 ft from EGL in all the borehole locations. After that stiff clay in most of the locations have been found along with 10-15 ft thick compressible black organic layers in BHs 1, 2, 4, 5 and 6. Hard clay/ fine sand with good bearing capacity have been found after 50-ft depth from EGL.

Foundation options

In this situation, the foundation should be selected considering the dimension, design load for the proposed building and bearing capacity of the soil. Deep foundation (preferably cast in-situ pile) may be selected as the most suitable option in this case. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found from the attached tables (8-11) in the report which is provided based on the test values provided.

Safety provisions

From the practical viewpoint, it has been found in many buildings that the structure failed due to soil settlement or judging the soil parameters incorrectly, providing no signs of structural damage. It is safe to consult a foundation (geotechnical) specialist before deciding on the foundation type, size and depth. The earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of a foundation.

Attachments

The layout of the boreholes of the investigated site and the bore-log profiles with SPT values are shown in the Appendix. Client performed no special laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative and the selection of type, depth and dimensions of foundation would be decided by the Geotechnical Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition, which is up to **60/65 ft** as the client provided.

Test Assisted By	Report Prepared By	Countersigned By
Md. Nazrul Islam	**Dr. Shriful Islam**	**Prof. Dr. H.M.A. Mahzuz**
Senior Technical Officer Dept. of CEE, SUST-3114	Associate Professor Dept. of CEE, SUST-3114	Coordinator, CRTC Dept. of CEE, SUST-3114